

Piloting Edge Copilot

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Piloting Edge Copilot - Speaker Deck

Piloting Edge Copilot

Code Blue 2024 presentation

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Jun Kokatsu

November 14, 2024

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Transcript

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Piloting Edge Copilot Jun Kokatsu

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self.origin • Former member of Microsoft Edge security team. •

Currently in Web security team at Google. • Bug hunter for 10 years. • @shhnjk

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What is Edge Copilot? • Copilot on Edge sidebar. •

It has access to contents on the active tab. • Many other privileged APIs are exposed and tightly integrate with Edge.

-

Architecture • edge://discover-chat WebUI has access to privileged APIs. •

Copilot UI is hosted in edgeservices.bing.com. • Communications between the WebUI and the iframe happens via postMessages.

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What is edge://discover-chat WebUI A browser internal page, with special

capabilities such as: • Access to camera and microphone by default. • Various public and private extension APIs: authPrivate, bookmarks, collectionsPrivate, history, metricsPrivate, search, tabGroups, tabs, windows. • Special Mojo interfaces to interact with websites and the browser, such as edge.copilot.mojom and underside_chat.mojom.

-

Security of edge://discover-chat SPA with strong CSP and Trusted Types,

effectively eliminating XSS. Content-Security-Policy: frame-src
https://edgeservices.bing.com/edgesvc/shell; require-trusted-types-for 'script'; script-src
edge://resources 'self'; frame-ancestors 'none'; trusted-types 'none';

Security of edgeservices.bing.com • Strict CSP (nonce and strict-dynamic). •

Trusted Types with policy enforcement (~10 custom policies). • Endpoint/origin based CSP allow-list for frame-src, connect-src, image-src, style-src, media-src. ◦ default-src 'self' for the rest. • Minimum CSP requirement enforced by CSP Embedded Enforcement (i.e. csp attribute in iframe). • Origin Isolation by the browser (per edge://process-internals/#site-isolation). ◦ Protects the origin from a renderer exploit triggered from other subdomains in bing.com.

What is CSP Embedded Enforcement? A mechanism to enforce a

minimum CSP restriction on iframe using csp attribute. For the iframe to render without an error, it must: 1. Return the same or stronger CSP header than the CSP defined in the csp attribute. or 2. Return Allow-CSP-From header to apply the minimum CSP restriction. a. e.g. Allow-CSP-From:
https://example.com

None

Nested frames to edgeservices.bing.com

CSP Embedded Enforcement

Summary • XSS seems impossible with Strict CSP and Trusted

Types on both edge://discover-chat and edgeservices.bing.com. • CSP Embedded Enforcement delegates to all nested iframes. • Seemingly no way for an attacker page to get a reference to the Edge Copilot sidebar. ◦ Can't open edge: URLs from normal websites ◦ Service worker, storages, etc, are double keyed. • Sh*t, it's secure.

Ignore the boring (secure) stuff, focus on interesting stuff Edge

Bing

Looking into www.bing.com Bing chat had a message listener where

it assigned message value to the iframe's src. `handleLoadFullScreenIframeEvent(O) { var B; this.config.features.enableFullScreenIframe && (this.fullScreenIframeUrl = O.url, null === (B = this.fullScreenIframeDialogRef) || void 0 === B || B.showModal()); }`

-

XSS on www.bing.com Sending javascript: URL via postMessage triggers XSS!

-

Edge exposes private API to Bing Following private APIs were

exposed to www.bing.com 🤖 • `chrome.edgeSplitTabsPrivate` • `chrome.edgeMarketingPagePrivate` • `chrome.edgeNurturingPrivate` • `chrome.edgeWalletDonationPrivate`

-

chrome.edgeSplitTabsPrivate Allows you to control split tabs in Edge.

-

chrome.edgeSplitTabsPrivate Allows you to control split tabs in Edge. Popup

blocker bypass: `chrome.edgeSplitTabsPrivate.openUrl( {"url":"https://www.example.com", "target":"SPLIT_TAB"}); chrome.edgeSplitTabsPrivate.exitSplitMode();`

-

chrome.edgeMarketingPagePrivate As the name suggests, some marketing related APIs. Send

arbitrary prompts to Edge copilot!! `prompt = "hello!"; chrome.edgeMarketingPagePrivate.sendNtpQuery( prompt, prompt, "https://www.example.com", e=>console.log(e));`

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How do we get an arbitrary site's content 1. XSS

on Bing. 2. Open an arbitrary website with popup blocker bypass. 3. Trigger Edge copilot with an arbitrary prompt. 4. ?

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How do we get an arbitrary site's content 1. XSS

on Bing. 2. Open an arbitrary website with popup blocker bypass. 3. Trigger Edge copilot with an arbitrary prompt. 4. ? Maybe ask copilot to summarize the page content, which should be available to Bing via chat history?

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Privacy feature blocking history syncing of web content

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Page intent detection by AI

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How Copilot knows about a site content? Site contents are

added as a message to the Edge copilot discussion.

-

The “bypass” 1. Ask copilot something unrelated to the page

(e.g. “Hi!”). 2. The AI decides not to flag for privacy (the chat is not related to the page). 3. Copilot still adds the site content to the history anyways 🤖

-

Demo <https://youtu.be/Vt75OIH7Iil>

-

One day, as I was browsing...

-

One day, as I was browsing...

-

One day, as I was browsing... document.title causes XSS

-

How? • Edge WebUI sends postMessage whenever title of the

page changes. • The message listener on Bing injects title as HTML. • While Trusted Types was enforced, pass-through policy was used for this code path. `createHTML(): s => { // No sanitization is performed
return s; }`

-

Still just an HTML injection... What to do?

-

Still just an HTML injection... What to do?

-

Still just an HTML injection... What to do? Permission Delegation

to Bing iframe!

-

Permission Delegation? • Permissions obtained by the top-level page can

be delegated to a cross-origin iframe using an allow attribute. • As explained, Edge WebUI has camera and microphone by default 😊 • An HTML injection can abuse this to delegate permissions to arbitrary sites. • Win?

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CSP frame-src 🙄

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Missing the last chain • CSP Embedded Enforcement delegates to

all nested iframes. ◦ All framable endpoints have very restrictive CSP (and almost always Strict CSP). ◦ Even there is an XSS on a framable endpoint, CSP would still block a script execution. • A few www.bing.com endpoints are framable, and I have a postMessage XSS on www.bing.com.

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HTML payload in title

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A link and a Bing iframe are injected Strict CSP

enforced on all iframes

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Clicking the link opens an attacker's page in a new

tab

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The attacker page gets opener reference to sidebar

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Triggers postMessage XSS on Bing

-

Access microphone through the opener reference!

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Demo <https://youtu.be/7NydJCndmws>

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A secret door to Edge Copilot • Any site could

embed edgeservices.bing.com. • But all privileged API and information were coming from edge://discover-chat. • What can we do with just embedding?

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A hashchange event listener • In addition to a message

listener, edgeservices.bing.com has a hashchange event listener. • It was acting as a command listener with the syntax of sjevt|{command}|{arguments}

-

Direct Prompt Injection • One of the command was “Discover.Chat.Say.User”,

which allows sending prompt to copilot on behalf of the user. ◦ #sjevt|Discover.Chat.Say.User|Hello! • How can we abuse this bug?

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Accessing Copilot’s memory When the copilot is asked about past

conversations, relevant past conversations are extracted and provided to copilot.

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Accessing Copilot’s memory When the copilot is asked about past

conversations, relevant past conversations are extracted and provided to copilot. How can we leak this past conversations?

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Exfiltration through Markdown • The most common way to exfiltrate

information from LLMs is through markdown images (i.e. [image: title]).

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Exfiltration through Markdown • The most common way to exfiltrate

information from LLMs is through markdown images (i.e. [image: title]). ◦ However, edgeservices.bing.com has image-src CSP :(

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Conclusion • Integrating a relatively secure system (Edge) with an

insecure system (Bing) will most likely produce an insecure system. ◦ You need to secure both of them, especially if integration is very tight. • AI specific risks matter, but classic application/browser security matters more. ◦ If there is an XSS, it can do almost anything on behalf of a victim. • Even if many of classic Web application security mitigations are deployed, attacks which uses AI-related exfiltration techniques are hard to mitigate.

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